

Quantum-Inspired AI for Reasoning: Beginner Guide

1) What this project is

This project improves how a small AI model solves multi-step questions (like math word problems and logic questions).

- Base model: small language model (~3B class)
- Main idea: generate many possible reasoning paths, then use optimization to pick the best subset
- Goal: beat normal prompting methods and target about 2x reasoning gain over baseline

2) Problem with older methods

Older methods in the papers work, but each has limits:

Method	Strength	Limitation for our goal
CoT (Wei et al., 2022)	Improves step-by-step reasoning	Uses one chain; error propagation
Self-Consistency (Wang et al., 2023)	Multiple chains + majority vote	Expensive; voting can keep wrong but common answers
ToT (Yao et al., 2023)	Strong search	High compute cost
CR/QUBO (Esencan et al., 2024)	Optimization-based reason selection	Built for large models; large QUBO size (~900 vars)
QCR-LLM (Flores-Garrigos et al., 2025)	Quantum/HUBO extension	Hardware heavy; not focused 3B SLM constraints

3) Our method (simple explanation)

We use a 3-stage pipeline:

1. Sampling: model generates multiple reasoning options
2. QUBO optimization: score and select best non-redundant reason set
3. Final inference: feed selected reasons back to model for final answer

In plain words: "Generate many ideas -> filter smartly -> answer better."

4) What we improved vs research papers

Improvement we added	Shortcoming resolved	Why it matters
SLM-adapted lightweight QUBO (semantic clustering to ≤ 200 vars)	Prior CR scale too large for efficient SLM use	Makes optimization practical on CPU/A100
Correctness-aware verifier term in QUBO	Similarity-only QUBO can keep fluent but wrong reasons	Reduces hallucinated/wrong chains
Diversity sampling with adaptive temperature + perturbation	SLM outputs can collapse to similar paths	Gives optimizer real choice diversity
QUBO-based hyperparameter search	Grid search is slow and local	Joint tuning improves quality/latency
Planned HUBO + optimized annealing	Pairwise interactions miss higher-order logic	Better global selection quality
QUBO-guided SFT loop (planned)	Inference-only gains plateau	Converts inference gains into model gains

5) Estimated improvement vs baselines

GSM8K projected cumulative path

Stage	Method	Estimated Accuracy
Baseline	Greedy SLM	~60%
+CoT	Standard prompting	~66%
+Diverse sampling	Better candidate generation	~72%
+QUBO selection	Combinatorial selection	~78%
+Verifier in QUBO	Correctness-aware selection	~84%
+SFT on selected traces	Feedback training	~88%
+Optimized annealing + HUBO	Advanced optimization	~91%

Interpretation: baseline to final is about +31 points (60 -> 91), which exceeds the practical 2x gain target definition used in roadmap notes.

6) Research papers identified from notes

1. Liu et al. (2025) - Understanding LLMs
2. Wei et al. (2022) - Chain-of-Thought Prompting
3. Kojima et al. (2022) - Zero-Shot CoT
4. Yao et al. (2023) - ReAct
5. Wang et al. (2023) - Self-Consistency
6. Yao et al. (2023) - Tree of Thoughts
7. Wang et al. (2025) - Ranked Voting Self-Consistency
8. Esencan et al. (2024) - Combinatorial Reasoning (QUBO)
9. Flores-Garrigos et al. (2025) - QCR-LLM
10. Chandarana et al. (2025) - Runtime Quantum Advantage
11. Margapuri et al. (2025) - PEPS+PPO

7) Why this is different from other methods

- Not just vote-based aggregation: uses optimization-backed subset selection.
- Not only large-model focused: explicitly engineered for smaller 3B-class models.
- Adds correctness signal directly into objective (key novelty).
- Targets deployable path now (CPU/A100), with future one-switch migration to quantum backend.

8) Future integration plan

- Integrate full multi-benchmark runner (GSM8K, BBH, StrategyQA, ARC, MMLU)
- Add HUBO and counterdiabatic-inspired annealing
- Run QUBO-guided SFT rounds to create self-improving loop
- Compare against all paper baselines with same evaluation protocol

9) Quick presentation points (non-technical)

- We do not trust one AI answer path; we generate many.
- We score quality + remove repetition mathematically.
- We keep only the best reasoning pieces.
- That selected reasoning helps the same small model answer better.
- This bridges low cost (small model) and high reasoning quality.